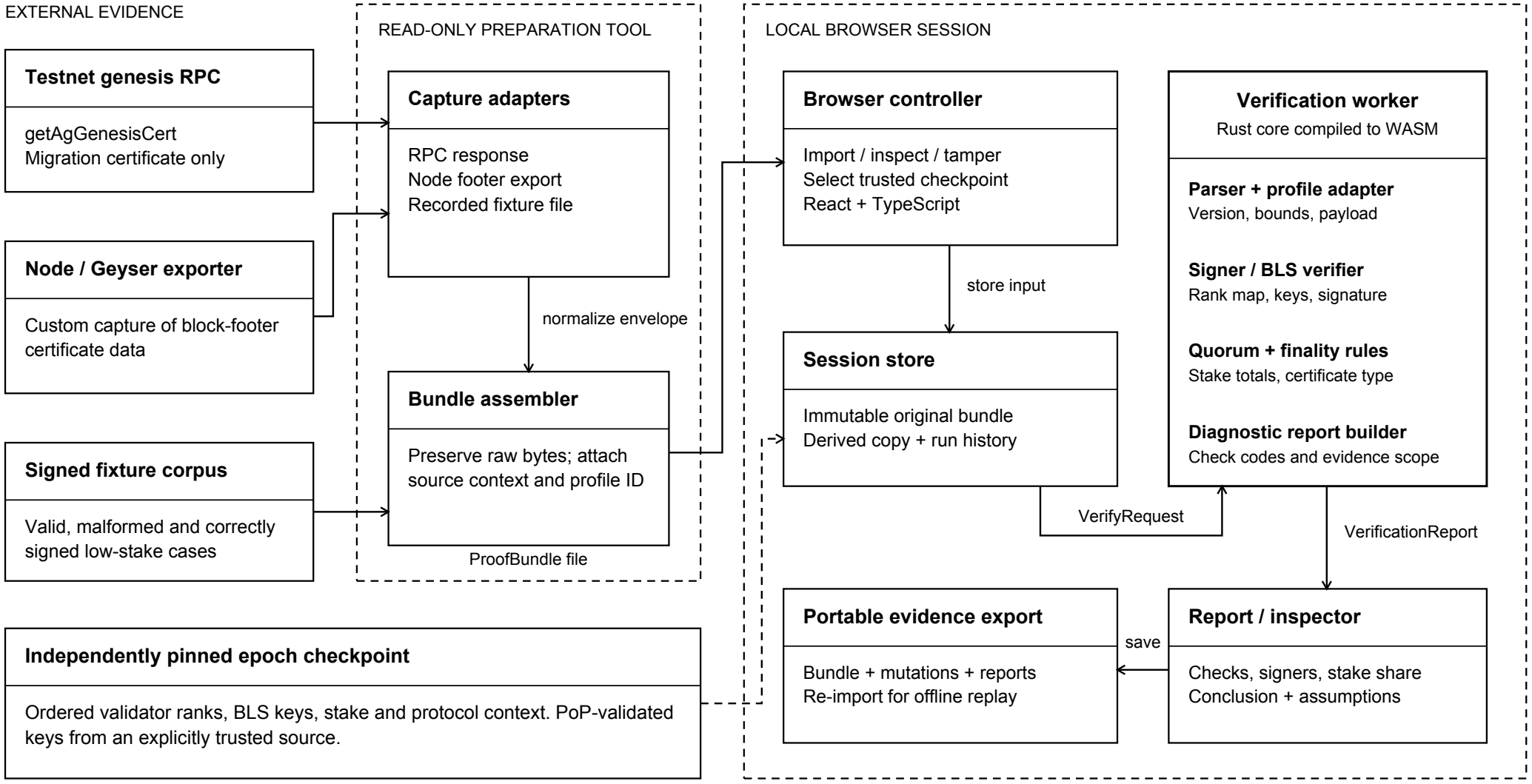
System components and data flow

ARCH-01

Browser-based certificate inspection, backed by one Rust verification core. All boxes below describe the proposed MVP.



Solid arrows: evidence and results. Dashed arrow: explicitly selected trust context. Dashed enclosures: runtime boundaries.

Solana: Alpenglow integration guide

Evidence bundle and verification contracts

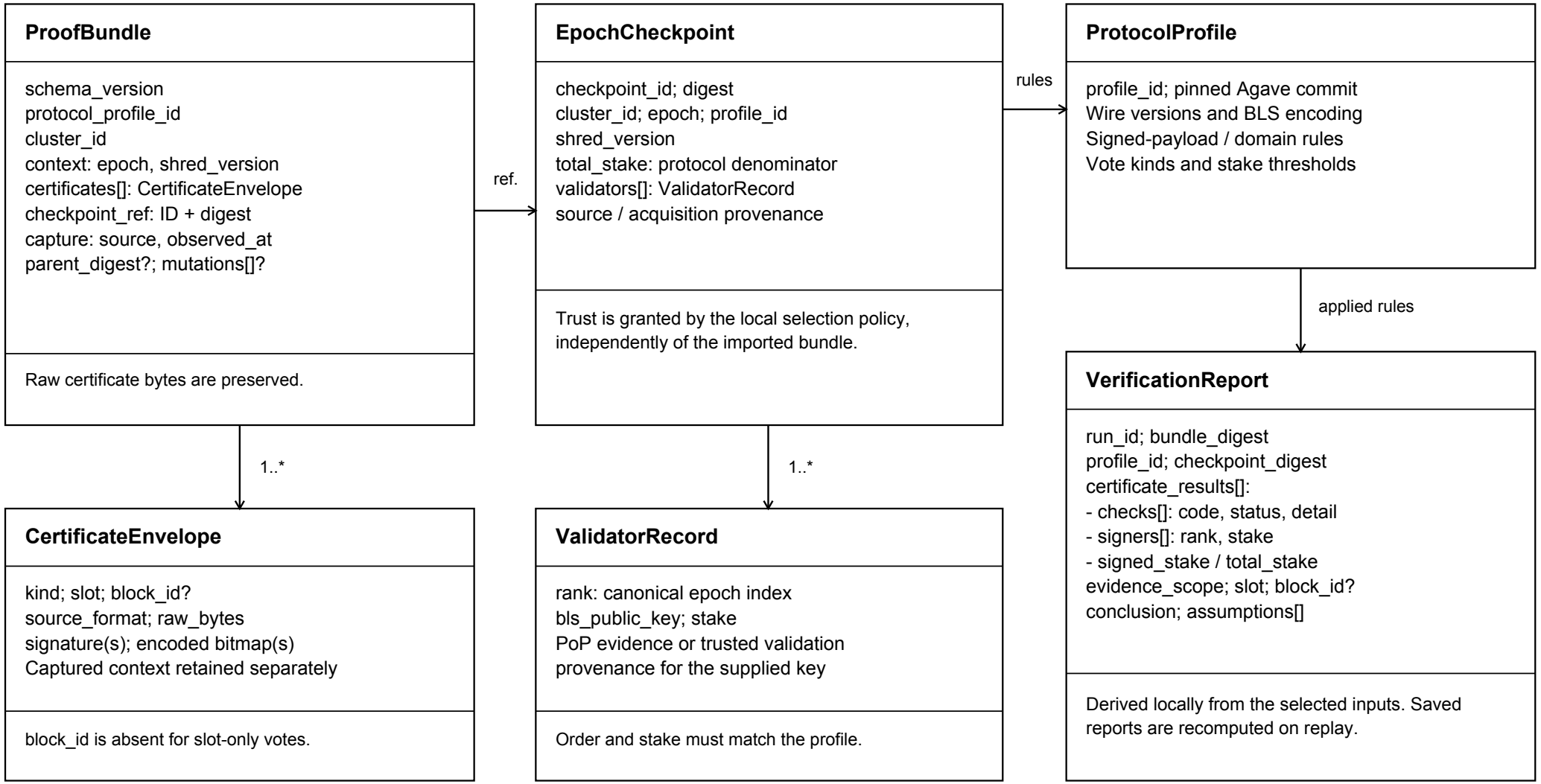
ARCH-02

Proposed versioned schemas. Imported metadata and file hashes identify evidence; they do not establish checkpoint trust.

A / PORTABLE INPUT

B / TRUSTED CONTEXT

C / RULES AND DERIVED OUTPUT

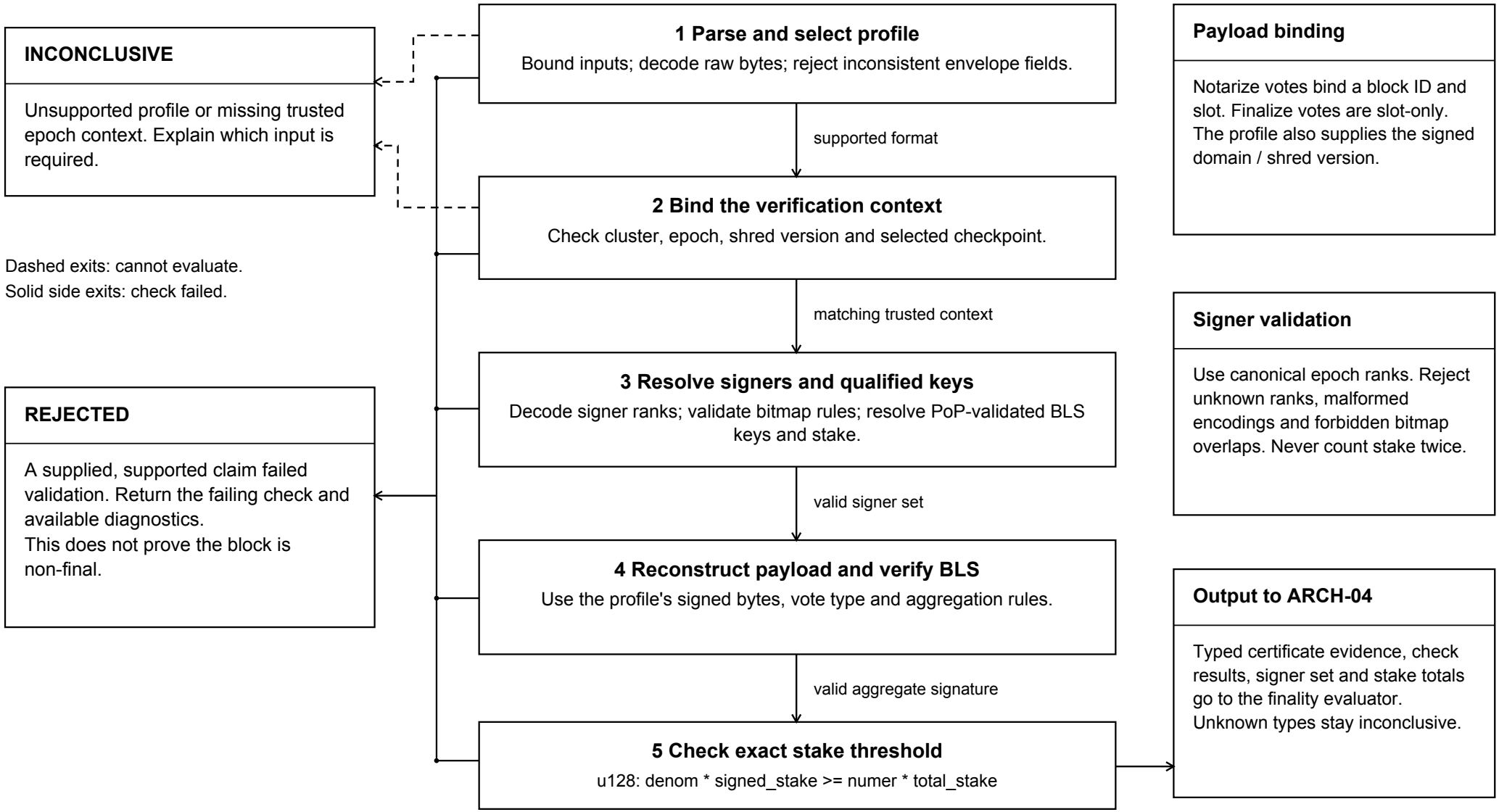


Agave: certificate verification

Certificate verification inside the Rust core

ARCH-03

Every run records per-check PASS / FAIL / NOT_CHECKED results. A valid signature alone does not establish block finality.



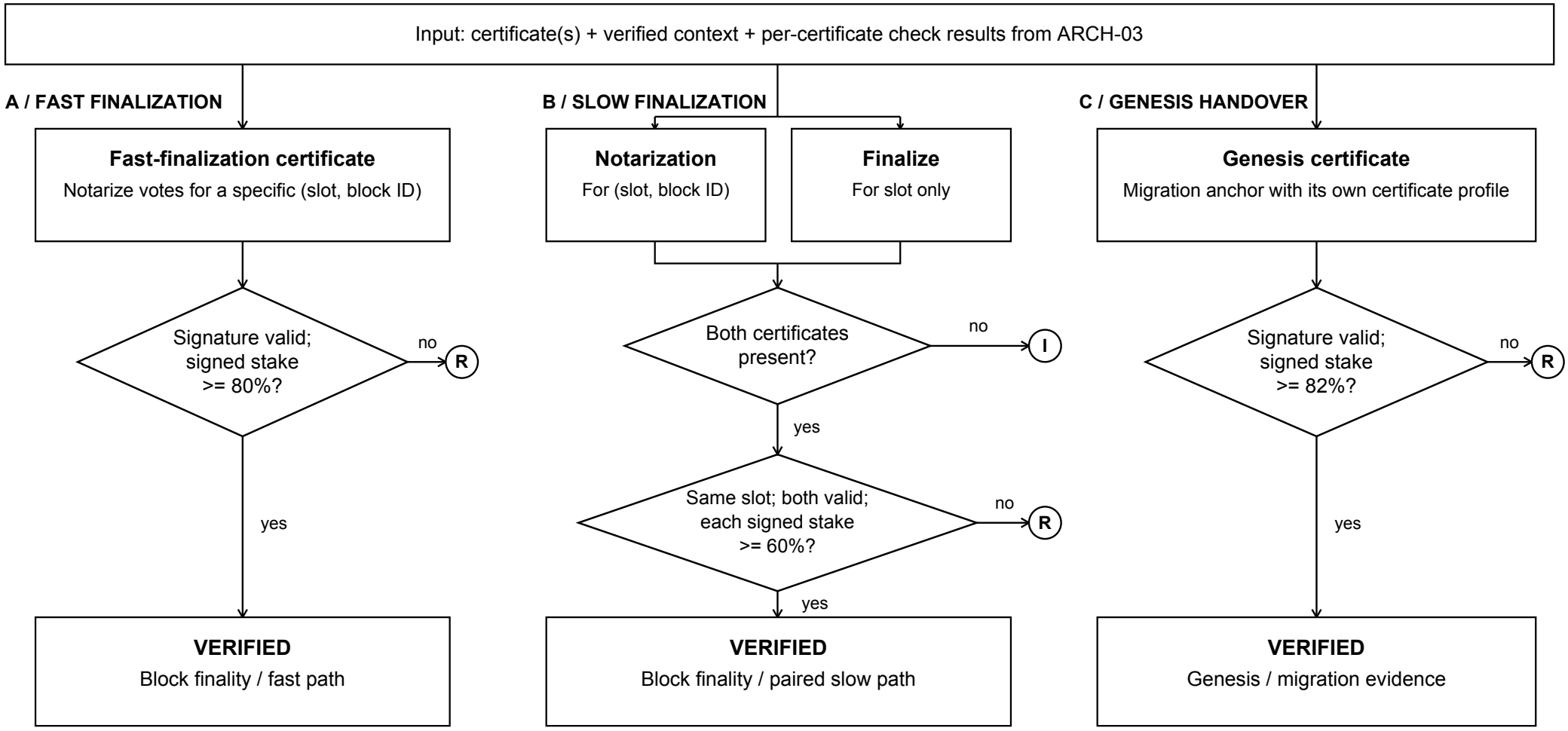
Dashed exits: cannot evaluate.
Solid side exits: check failed.

Agave: certificate verification

Finality evaluation and result semantics

ARCH-04

The evaluator combines typed certificate evidence and check results, always relative to the selected trusted checkpoint.



R = REJECTED: a supported claim fails a required check.
I = INCONCLUSIVE: evidence is incomplete or cannot be evaluated.

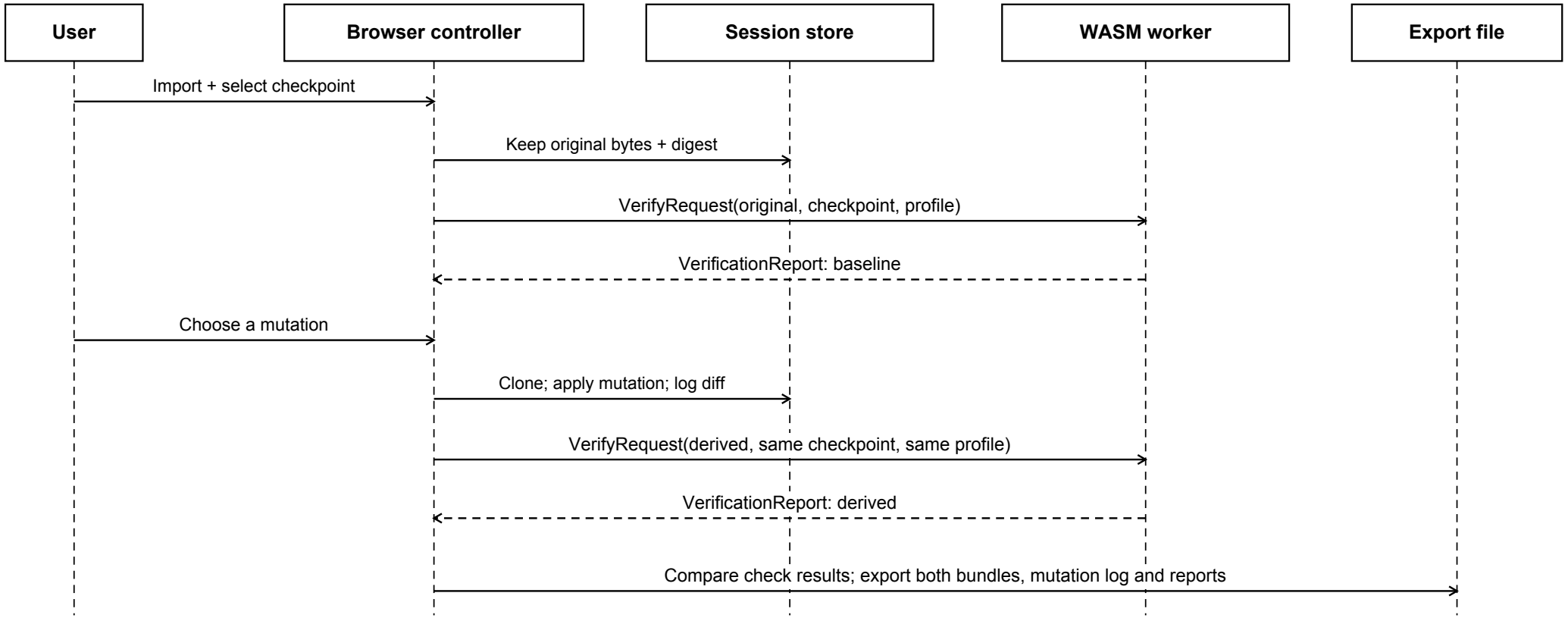
A notarization certificate alone does not establish finality. Genesis evidence is reported separately. Certificates do not attest to transaction execution results.

SIMD-0326: Alpenglow protocol Agave: migration / genesis certificate

Tamper lab, comparison and offline replay

ARCH-05

The same core verifies captured data and synthetic fixtures. A changed bundle always receives a new run and a fresh report.



Replay: import the saved evidence, independently select its checkpoint, then rerun verification. Stored reports are reference output.

Test input	Expected diagnostic	Finality conclusion
Alter a signed block ID or signature	BLS verification fails	REJECTED
Correctly signed under-quorum fixture	BLS passes; stake threshold fails	REJECTED
Remove the required trusted checkpoint	Context cannot be established	INCONCLUSIVE
Provide only a 60% notarization certificate	Missing finalization counterpart	INCONCLUSIVE